

### Introduction

#### The Microscope

Tissue slides are mounted on the stage. The on/off switch at the base activates the light source; the adjacent sliding switch is a rheostat that regulates the bulb's intensity. Light derived from the source in the base is directed upward beneath the stage through a condenser lens, which is fitted with an adjustable diaphragm. The narrow, bright point of light produced by this lens should pass through the aperture in the stage and focus on the tissue section. Generally, the condenser lens is adjusted when it is near the top of its run.

#### Objective Lenses

The illuminated image is transmitted upwards and passes through the objective lens. The microscope is outfitted with a turret with 4 lenses: a low power 4X (red ring), an intermediate 10X (orange ring), a high power 40X (green ring) and a higher-powered oil immersion 100X (blue ring). The selected objective lens is rotated so that it lies directly above the condenser lens.

The transmitted image then passes into the head of the microscope and through the 2 eyepieces. The two oculars can be either pulled apart or pushed together to accommodate individual differences in interpupillary distances. When out of adjustment, you may not be able to converge the images directed toward each eye.

#### Types of Microscopy

**Light** (bright field) is the most common type of microscopy and is used to observe stained sections. Note that different stains highlight different shapes and morphologies but may obscure more familiar structures.

**Phase Contrast** is mainly used to observe living cells, often in cell culture dishes. Cells are visible as a black and white image; magnification and resolution of details is limited in comparison to the light microscope.

**Nomarski** (differential interference) is similar to phase contrast but provides greater resolution of the 3-dimensional organization of the cells or tissues.

**Dark Field** illuminates cells and tissues against a dark, contrasting background. It is used with techniques like *in situ* hybridization or autoradiography, each of which localizes a molecule of interest within a cell or tissue.

**Transmission Electron Microscopy (TEM)** magnifies the image so that one may visualize intracellular structures and organelles. Tissue processing and staining are quite different for both types of electron microscopy.

**Scanning Electron Microscopy (SEM)** scans cell surfaces at great magnifications providing an image of highly resolved surface details.

**Scanning Force Microscopy (SFM)** scans cells, parts of cells, macromolecules (membrane, membrane channels, DNA, RNA, etc.) with computerized imaging (image metal grafts and

macromolecules implants) and **Atomic Force Microscopy (AFM)** scans thin layers of molecules and can visualize heavy atoms, mainly used in physics but with promises for biology (metal-biotech).

## **Tissue Preparation**

Biological tissues are preserved by chemical fixatives (e.g., formalin, ethanol, glutaraldehyde) that cross-link and precipitate the organic molecules. The tissues are then dehydrated using organic solvents. The solvents are replaced by either paraffin or plastic resins which embed the former soft material in a hard matrix which facilitates making thin sections (e.g., it is easier to slice a thin section of frozen meat than a thawed piece). For light microscopy, sections range from 1 - 10 microns in thickness.

**Fixation** prevents autolysis of cells and preserves their morphology. It may take hours to weeks depending on the tissue size and density.

After fixing, tissue samples are **dehydrated** in 100% alcohol and cleared in a solution (xylene) that can mix with both the alcohol and the embedding medium (paraffin wax).

**Embedding** involves infiltration with the embedding media (paraffin or plastic) to stiffen the tissue.

**Sectioning** is the cutting of very thin sections from a block of embedded tissue. Paraffin sections are cut at 5 - 10 microns and then adhered to a glass microscope slide.

**Staining** of sections is done after removing the embedding medium from the tissue by solubilizing it in a clearing agent. The tissue is then re-hydrated through graded alcohol rinses to water. Sections on the slides are stained in aqueous solutions and washed in water to remove unbound stain. Different stains react with different biochemical components of cells, highlighting different morphological features and imparting a colorful appearance to the tissue.

**Mounting** preserves the section by placement of a coverslip to the slide. The coverslip has a similar refractive index that maintains a high degree of transparency.

In biological tissues, the basic molecular components (nucleic acids, proteins, saccharides and lipids) have different cytological appearances based on the fixatives and histochemical stains used to prepare the tissue.

## **Differential Staining Properties**

Sections are stained with dyes to make them visible in the light beam. The two most widely used stains are hematoxylin and eosin (H&E). While other stains are used in your preparations, we will focus primarily on the H&E, though other stains will be described when they are employed.

### **Hematoxylin and Nucleic Acids**

Hematoxylin is a blue-black dye that has a basic pH. Since objects are stained with oppositely charged dyes, hematoxylin should preferentially bind acid molecules. Consequently, nucleic acid associated molecules (DNA and RNA) will stain from blue to black based on the density of the material. In normal preparations, nuclear DNA and cytoplasmic ribosomes will be darkly stained

by hematoxylin. Because they bind a basic dye these acidic molecules are basophilic (i.e., “a lover of bases”).

### **Eosin and Proteins**

Eosin is an acidic, red dye. Molecules with a basic pH will bind eosin; such molecules are referred to as being eosinophilic (acidophilic). Since proteins are amines, they tend to stain with acid dyes. Therefore, proteins stain pink to red in H&E preparations.

### **Intracellular and Extracellular H&E Staining**

If a cell contains a lot of protein (e.g., a muscle cell filled with actin and myosin), the cytoplasm will stain red. If the cell, however, is synthesizing large quantities of protein (embryonic cell, a protein secreting cell or a transformed cancer cell), its cytoplasm will possess many ribosomes. The blue-black of the ribosomes and the red of the proteins create a purple (basophilic) cytoplasm. Since nuclear chromatin is comprised of both DNA and protein (histone), nuclei exhibit both blue and red staining simultaneously. That is why they generally appear as purple in H&E sections. The extracellular fibrous materials are primarily collagen proteins; they are quite eosinophilic and stain bright pink to red.

### **Saccharide and Lipid Staining**

Saccharides and mucopolysaccharides are not readily observed in tissue sections because:

They do not preserve well with normal fixation procedures and are frequently washed-out.

They do not stain well with H&E, although acid mucopolysaccharides may exhibit a bluish tint.

Saccharides are generally visualized using special staining procedures such as periodic acid - Schiff (PAS) dyes. When used in this course, special saccharide stains will be discussed in the description of the slides.

Lipids are rarely observed in normal preparations because they are washed-out during dehydration with organic solvents. Use of special fixatives (e.g., osmium) and/or dyes (e.g., Sudan black) will retain and stain cytoplasmic lipids.

### **Artifacts**

Artifacts are distortions and aberrations of tissue structure that are produced when tissues are inadequately preserved or prepared. Common artifacts include shrinkage, clumping and swelling of cells, loss of lining epithelia, artificial tearing and separation of tissues and cells.

Be careful, such artifacts are frequently present in our slide sets, and they might influence your interpretation of tissue and organ structure.

### **Identifying Cells**

**Cells** are membrane bound units which, when aggregated in large numbers, form **tissues** (e.g. epithelial cells combine to form a tissue called epithelium). Different types of tissues, when

aggregated in a specific structure, form **organs** (e.g. the spleen is made of vascular tissue and connective tissue).

**Somatic cells** include all cells of the body except **germ cells** which are located only in the gonads. Somatic cells, generally, divide by mitosis into identical cells while germ cells divide by meiosis to produce non-identical gametes (i.e. sperm and ova).

Each type of cell (a muscle or nerve cell, for example) has different bio-chemical and physiological functions. To accomplish their different functions, these cells have different morphological structures and their microscopic appearances vary enabling us to distinguish different cell and tissue types.

Cells and tissues can be seen in the light microscope, but an electron microscope is needed to see most of the intracellular structures / organelles.

**Nucleus:** The most obvious and discrete structures observed in any H&E stained section are the densely stained nuclei. These “spherical” structures are generally suspended within the center of the cell’s cytoplasm. The rounded contours of the nuclear envelope are enhanced and delimited by dark-staining chromatin (blue-black to dark purple) that is bound to the inner nuclear envelope.

Since the nucleus is suspended within the cytoplasm, its shape usually corresponds to the overall shape of the cell. This is an important point, for it is frequently difficult to distinguish the peripheral margins of many cells.

Therefore, the nucleus can be used as an indicator of a cell’s morphology. Cuboidal cells have round nuclei; columnar cells have an elongated bean-shaped nucleus; and squamous cells have flattened nuclei that resemble the yolk in a fried egg.

**Cytoplasm:** The other cytoplasmic organelles (e.g., endoplasmic reticulum, Golgi apparatus, mitochondria, microtubules) are either below the resolution of the light microscope or are not visible without the use of special preparations. However, the collective staining of these constituents imparts distinctive coloration which enables one to visualize the extent of the cell.

## Laboratory Experience

As part of the exercise of this lab, you will learn to recognize individual cells by observing their nuclei and cytoplasmic appearance and you will practice presenting slides to your colleagues.

In the following slides, concentrate on the cells mentioned noting their similarities and their differences. Note how the structure of the cells can be indicative of the function of the cells. All these tissues will be examined in more detail later in the course.

### Thyroid, H&E: Slide # 38-40

- The thyroid gland is in the front of the neck. The gland is covered by a connective tissue capsule. Within this capsule, the cells that secrete the hormone are organized as follicles. These spherical follicles are lined by a **simple cuboidal epithelium** (box-like cells; characterized by the nuclear shape). The lumen is filled with a secretion product (colloidal substance containing thyroglobulin).
- The apical surface of the epithelial cells faces the colloid; the basal surfaces sit upon a matrix of extracellular collagen fibrils. In many portions of the slide the epithelial cytoplasm is not preserved and not stained (an artifact).

- Outside of the follicles are blood vessels filled with non-nucleated, erythrocytes. The internal lining of the vessel is comprised of a flattened **simple squamous epithelium** called an endothelium.
- These follicles are the functional unit of the thyroid gland. Therefore, they are called the **parenchyma**. The supporting connective tissue around the follicles is termed **stroma**.

#### **Uterus, H&E: Slide # 41-43**

- The uterus is a pelvic organ. It is pear shaped and has a cavity within called the uterine cavity. The image provided to you shows the epithelial cells lining the uterine cavity, under low magnification so that you can appreciate how glands are formed by invagination of epithelium. In a higher magnification image, you can appreciate the type of cells.
- Along the luminal surface you can observe an epithelium comprised of **simple columnar epithelium** (single layer of tall cells). There is more euchromatin in the elliptical shaped nuclei than in the denser, more heterochromatic nuclei of the cells beneath the basement membrane.
- The surface epithelium forms **tubular glands** that penetrate the underlying tissue. Because of the tangential section, most of the tubular glands cannot be followed down their full length. Rather, one sees elliptical and circular profiles of sectioned glands deeper in the tissue. Many of the glands show a clear “halo” between themselves and the surrounding tissue; this is a shrinkage artifact.

#### **Small Intestine (Duodenum), H&E: Slide # 44-46**

- Duodenum is a part of small intestine, which is a tubular tract, adapted to increase the surface area of absorption. Villi (finger-like projections of the connective tissue) extend into the lumen in this region of the small intestine. These villi are lined by a **simple columnar epithelium** containing three cell types:
- The dense purple stained columnar cells are **absorptive cells**. Their nuclei have a predominance of euchromatin and distinct multiple nucleoli. The vacuolated cells with the clear space are the **goblet cells** (resembles a wine goblet with a stem). Goblet cell nuclei, with condensed heterochromatin, are, generally, found in the “stem” of the cell beneath the clear space. Within the epithelium are numerous dense (heterochromatic), round nuclei that are usually surrounded by a clear halo. These cells represent small **lymphocytes** that have penetrated the epithelium as part of their immune activities. These migratory cells are not bound to the epithelium.

#### **Check List:**

- Identify the stain used in each slide.
- Identify cells and describe cell shape.
- Identify nuclei.
- Identify cytoplasm.
- Identify the stain generally taken up by the nucleus.

- Identify the stain generally taken up the cytoplasm.
- Define & identify stroma and parenchyma.

**Study Questions:**

A highly active cell contains a euchromatic or heterochromatic nucleus?

The cytoplasm of a cell that contains large quantities of protein would stain eosinophilic or basophilic?

Hematoxylin would preferentially stain chromatin or collagen?

Polysaccharides and mucopolysaccharides are best stained with eosin, Sudan black, hematoxylin or periodic acid – Schiff?